

Compiler Construction (CS-4031)

Final Exam

Date: 12/30/2025

Total Time: 3hrs

Course Instructors

Total Marks: 100

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Total Questions: 05

Roll No

Section

Student Signature

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Attempt all questions.

CLO # 1: Understand the basic techniques used in compiler construction such as lexical analysis, top-down, bottom-up parsing, context-sensitive analysis, and intermediate code generation.

Question 1:

(10 + 10 = 20 points)

a) Consider the following grammar:

$S' \rightarrow S$

$S \rightarrow id = E$

$E \rightarrow E + T$

$E \rightarrow T$

$T \rightarrow id$

- i. Find all CLR(1) items and construct CLR(1) parsing table.
- ii. Identify all the states which can be merged.

b) For each of the following programming language rules, name the compiler construction phase in which that rule is checked:

- i. A function call has the correct number of arguments.
- ii. Digits [0-9] may appear in identifiers, but not as the first character e.g., *a123* and *pi314* are valid identifiers, but *2big* is not.
- iii. Every variable must be declared before it is used in the program.
- iv. Assignments such as *a = 42;* must end with a semicolon (;).
- v. Array indices are integers within valid bounds.

CLO # 2: Understand the data structures used in compiler construction such as AST, symbol tables and three-address code

Question 2:

(10 + 10 = 20 points)

a) Following the Expressions Given, you are required to convert these into Three Address Code and Directed Acyclic Graph.

- i. $result = (a * b) + (c - d) * (a * b) / (e + f)$
- ii. $q = (a + b + c) * (a + b + c) - a * a - b * b - c * c$

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b) Briefly answer the following:

- i. List down the common forms of 3-Address Code.
- ii. Discuss the role of temporary variables in 3-Address Code.

CLO # 2: Understand the data structures used in compiler construction such as AST, symbol tables and three-address code

Question 3:

(10 + 10 = 20 points)

a) Consider the given grammar and its semantic rules.

PRODUCTION	SEMANTIC RULES
1) $S \rightarrow B$	$B.ps = 10$
2) $B \rightarrow B_1 B_2$	$B_1.ps = B.ps$ $B_2.ps = B.ps$ $B.ht = \max(B_1.ht, B_2.ht)$ $B.dp = \max(B_1.dp, B_2.dp)$
3) $B \rightarrow B_1 \text{ sub } B_2$	$B_1.ps = B.ps$ $B_2.ps = 0.7 \times B.ps$ $B.ht = \max(B_1.ht, B_2.ht - 0.25 \times B.ps)$ $B.dp = \max(B_1.dp, B_2.dp + 0.25 \times B.ps)$
4) $B \rightarrow (B_1)$	$B_1.ps = B.ps$ $B.ht = B_1.ht$ $B.dp = B_1.dp$
5) $B \rightarrow \text{text}$	$B.ht = \text{getHt}(B.ps, \text{text.lexval})$ $B.dp = \text{getDp}(B.ps, \text{text.lexval})$

- i. Identify which attributes are *synthesized* and which ones are *inherited*.
- ii. Draw an annotated parse tree for the following string:

(text sub (text sub text)) text

- iii. Starting with **B.ps = 10** at the root (from $S \rightarrow B$), compute the values for **ps**, **ht**, and **dp** for every **B** node in the tree. Show your steps for each internal node. Give the final values of **ps**, **ht**, and **dp** at the root **B**.

b) Identify all the semantic errors present in the following code snippet and also write the corrected code:

```

1.      int main() {
2.      int* p;
3.      cout << *p;
4.      int arr[3] = {1,2,3};
5.      arr[5] = 10;
6.      int x = 10, y = 0;
7.      int z = x / y;
8.      const int c = 5;
9.      int* q = (int*) &c;
10.     *q = 10;
11.     return 0;
12.     }
```

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CLO # 1: Understand the basic techniques used in compiler construction such as lexical analysis, top-down, bottom-up parsing, context-sensitive analysis, and intermediate code generation.

Question 4:

(10 + 10 = 20 points)

- a) Identify all the statements where machine-independent code optimization techniques can be used in the following three-address code. You must mention name of the technique along with the line number where it is used. Also rewrite the code after performing the optimization.

1.	t1 = a + b
2.	t2 = a + b
3.	t3 = t1 * 1
4.	t4 = 0 + t2
5.	t5 = t3
6.	t6 = t4 * c
7.	t7 = t5 * c
8.	t8 = t6 - t7
9.	t9 = d * 0
10.	t10 = t9 + e
11.	f = t10
12.	g = t1 + b

- b) Briefly answer the following:
- Machine-dependent vs machine-independent code optimization.
 - Directed acyclic graph vs abstract syntax tree.

CLO # 3: Design and implement a compiler using a software engineering approach.

Question 5:

(10 + 10 = 20 points)

- a) Convert the following C code into its corresponding three-address code and Assembly code.

1.	#include <stdio.h>
2.	int main() {
3.	int radius, i;
4.	int pi = 3;
5.	for(i = 0; i < 4; i++) {
6.	radius = i + 2;
7.	int area = pi * radius * radius;
8.	printf("Area: %d\n", area); }
9.	return 0; }

- b) In *NewC++* language, function names may start with a digit. Arguments are enclosed in brackets [] and separated by semi-colon ;. The second argument is optional and defaults to 0 if not provided.

Examples: 3foo[5;7]
1bar[9]

Write the regular expression, CFG and syntax-directed translation for these function calls.

- Good luck -